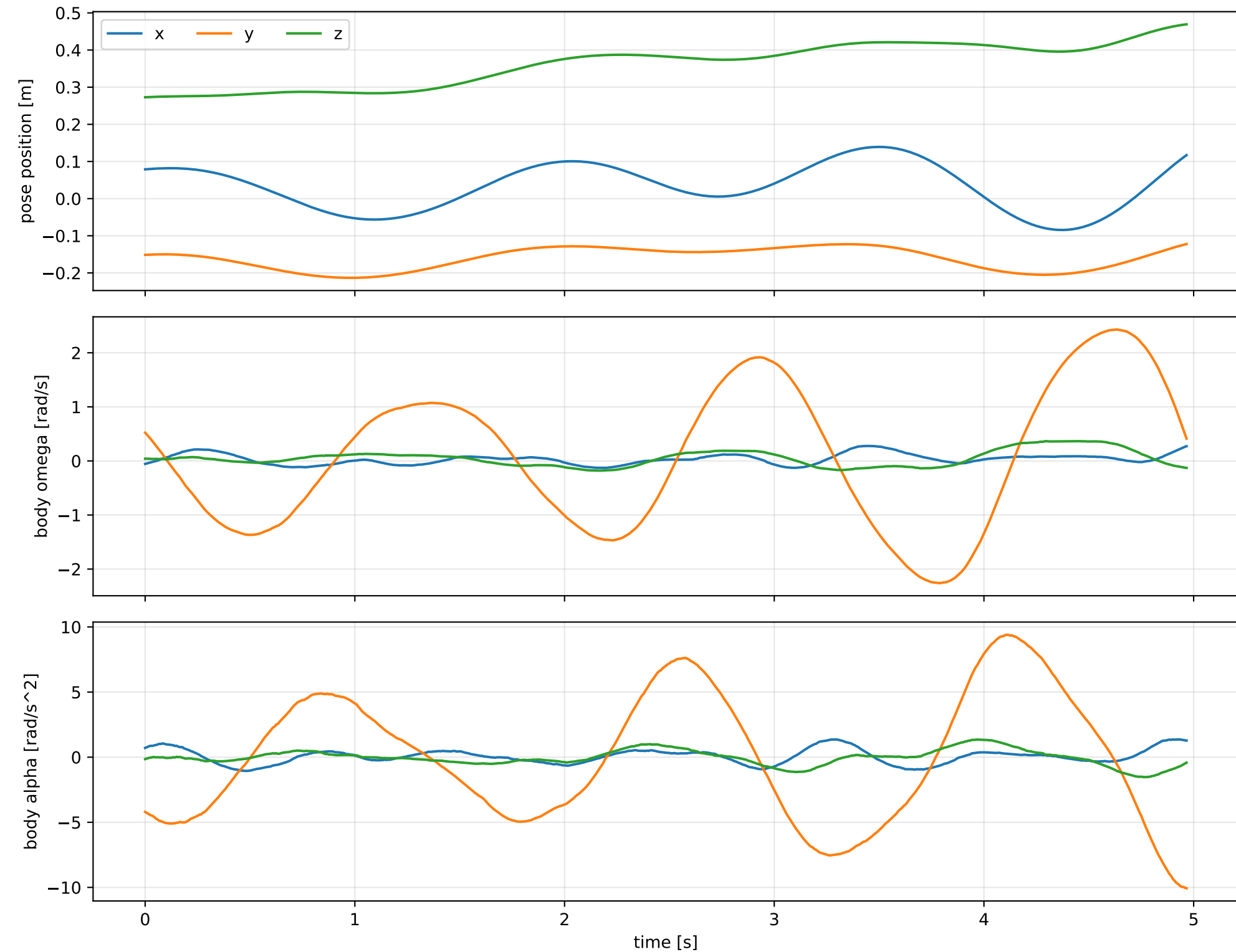
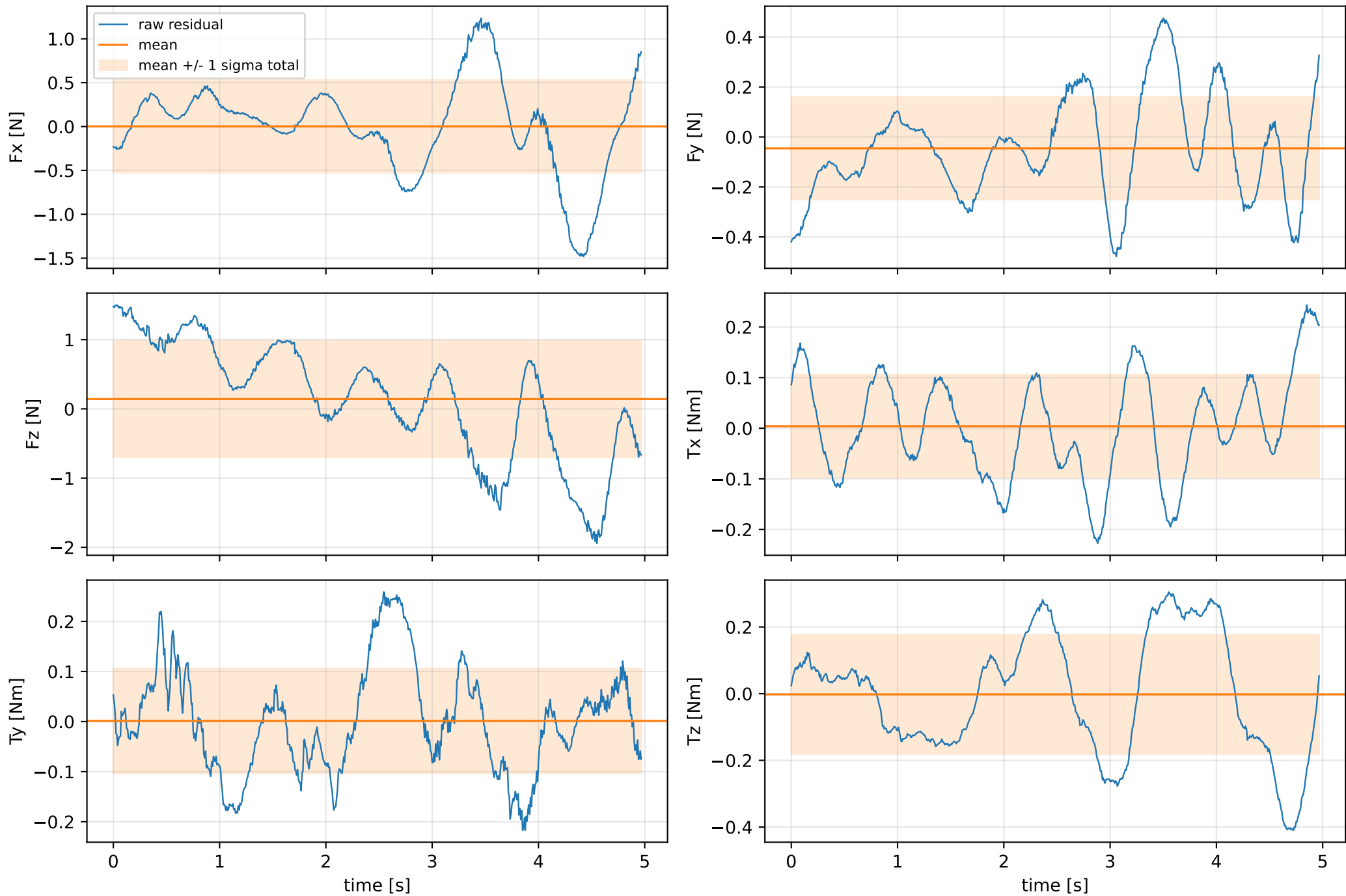


prior_free: SG trajectory and derived motion



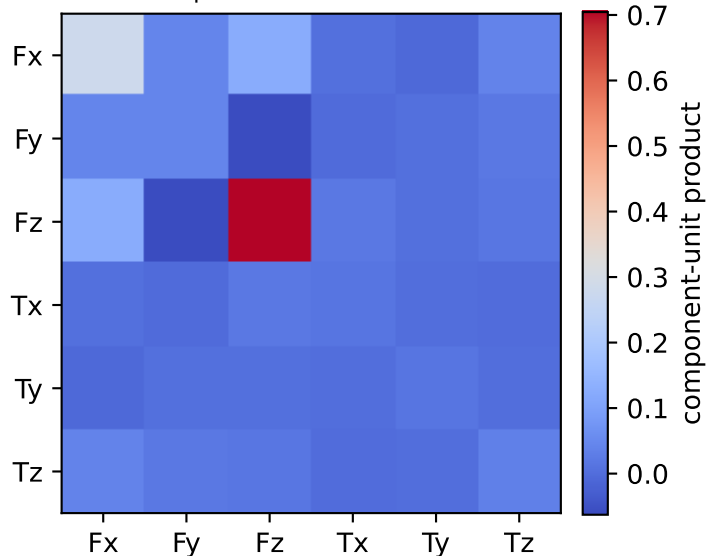
prior_free: raw Newton--Euler residual wrench (not trajectory-fitted external wrench)
mass gauge fixed to nominal mass = 2.35159759 kg



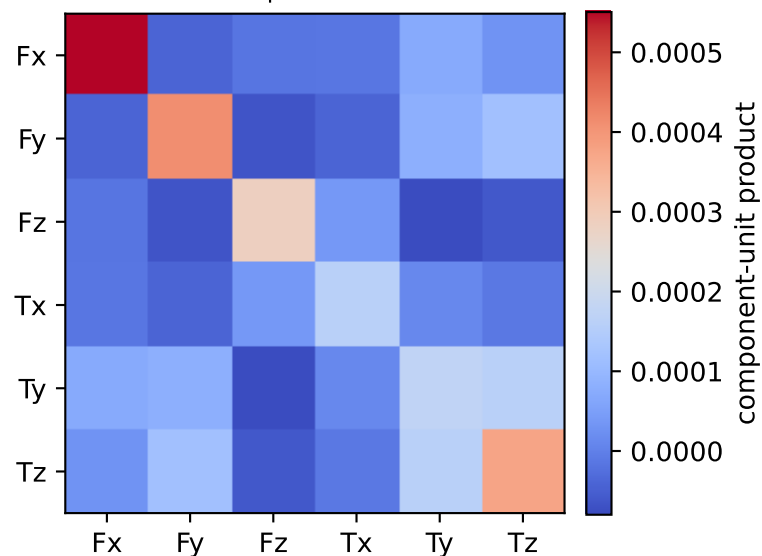
vector RMS about zero: force 1.0219 N, torque 0.2308 Nm; component std about mean: force [0.5268 0.2047 0.8393], torque [0.1014 0.1049 0.179]

prior_free: residual-wrench covariance decomposition in nominal-mass gauge
force [N], torque [Nm]; raw excess eigenvalues [0.008 0.009 0.016 0.035 0.266 0.741]

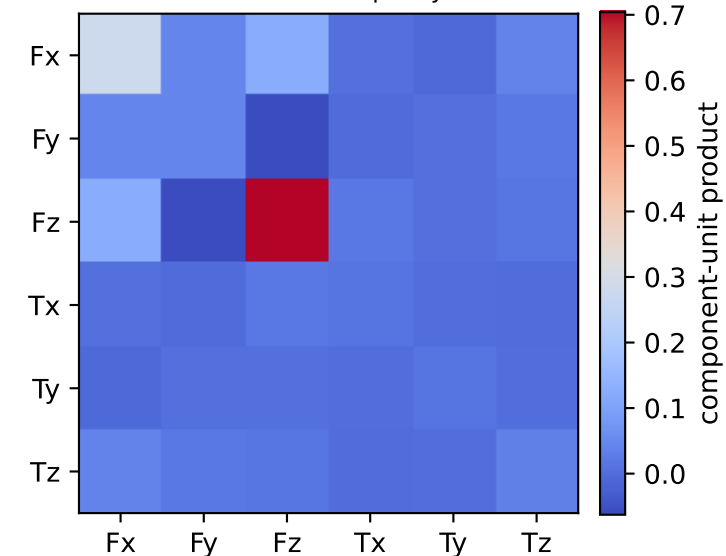
empirical total covariance



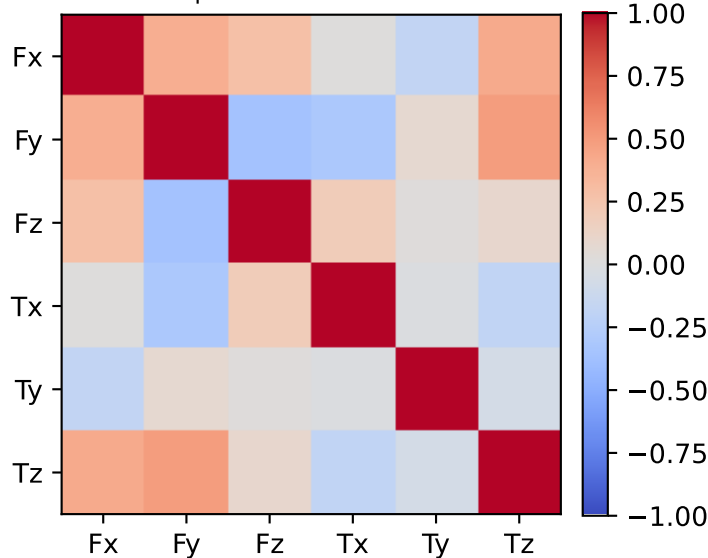
mean full-SG prediction covariance



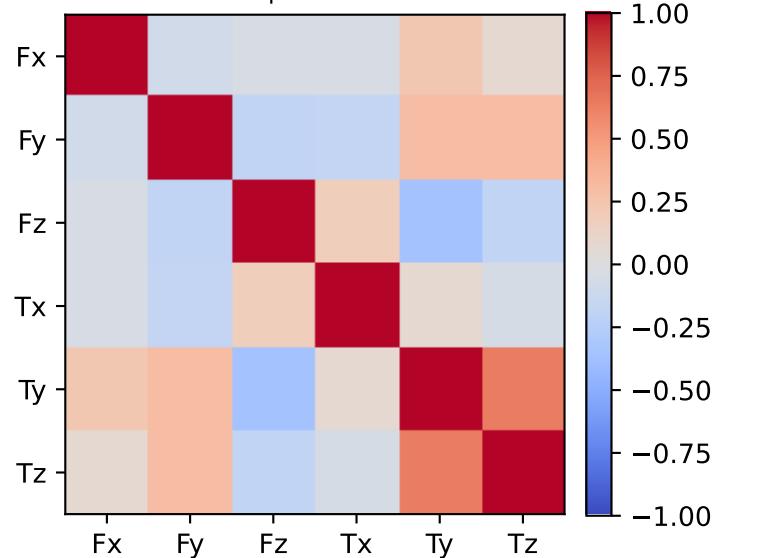
PSD excess model discrepancy covariance



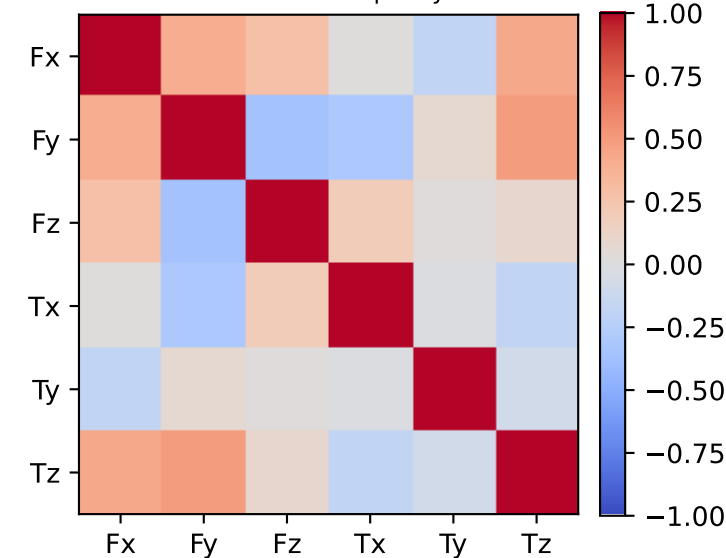
empirical total correlation



mean full-SG prediction correlation

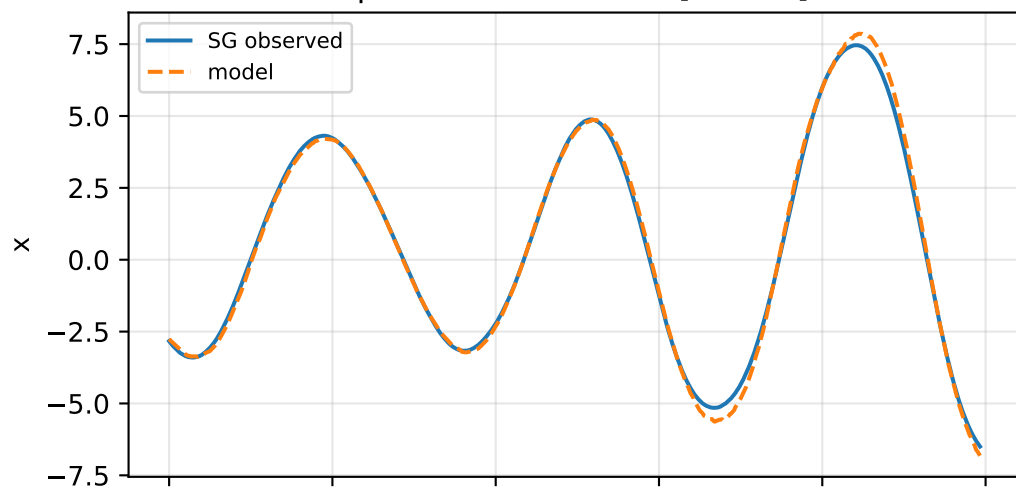


PSD excess model discrepancy correlation

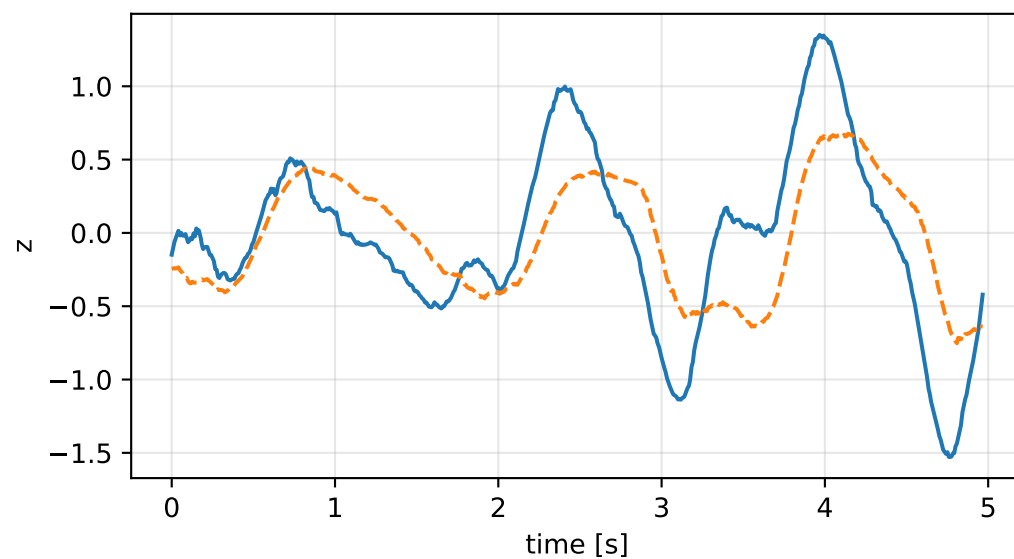
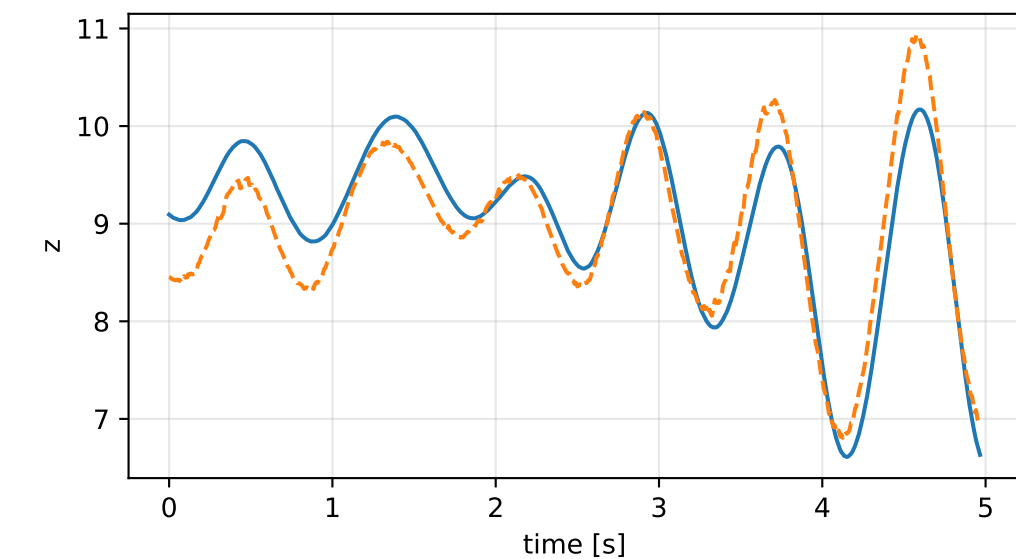
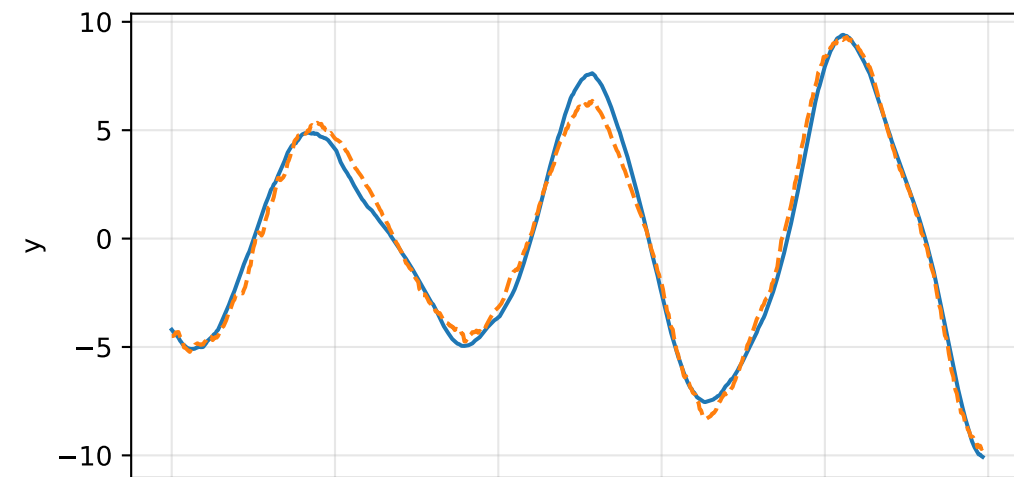
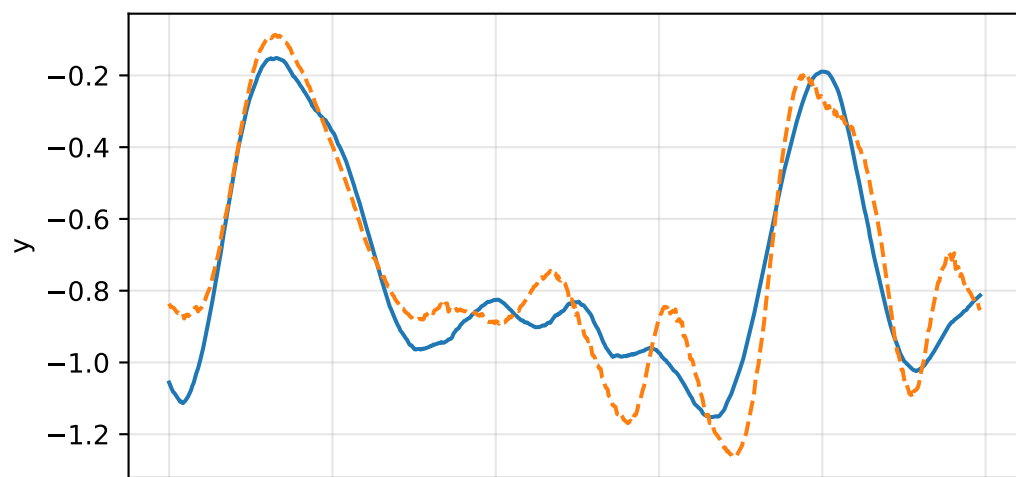
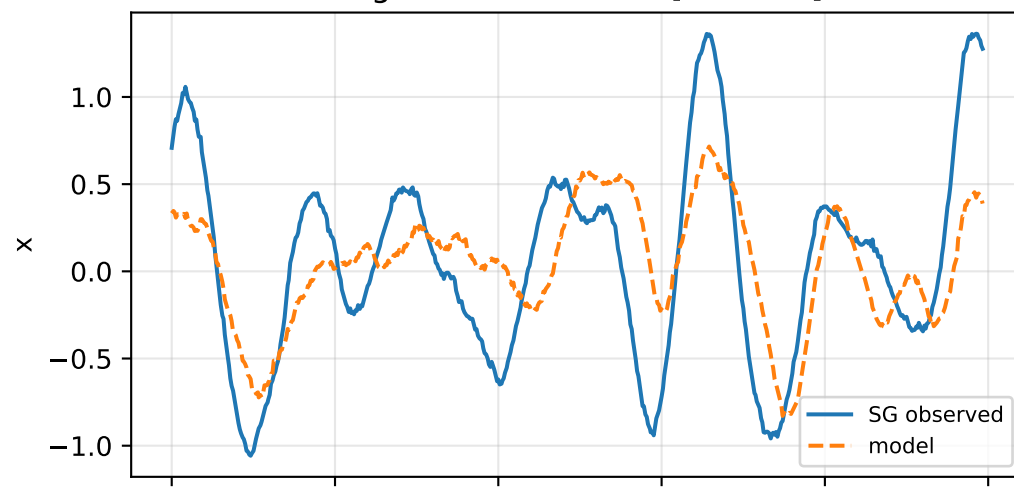


prior_free: acceleration residual objective

specific acceleration [m/s²]

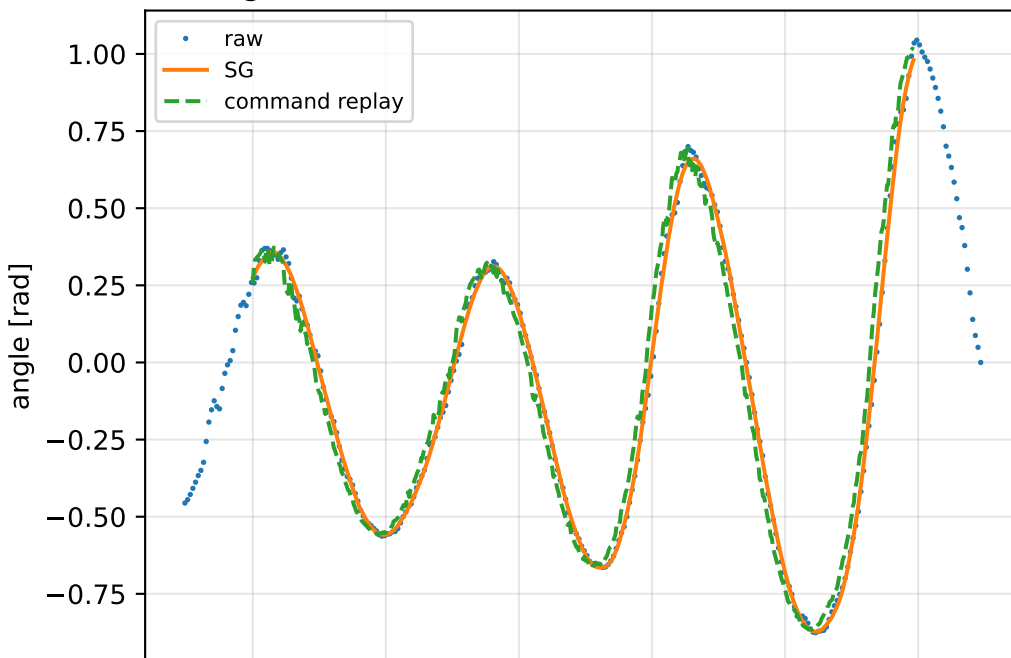


angular acceleration [rad/s²]

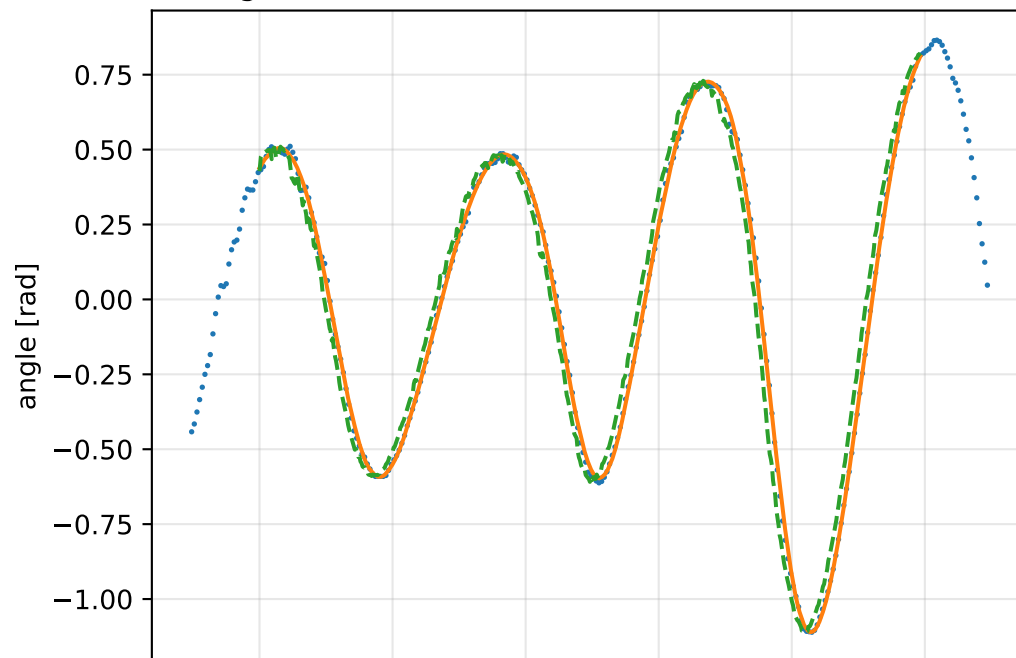


prior_free: gimbal measurement audit (objective=measured_sg)

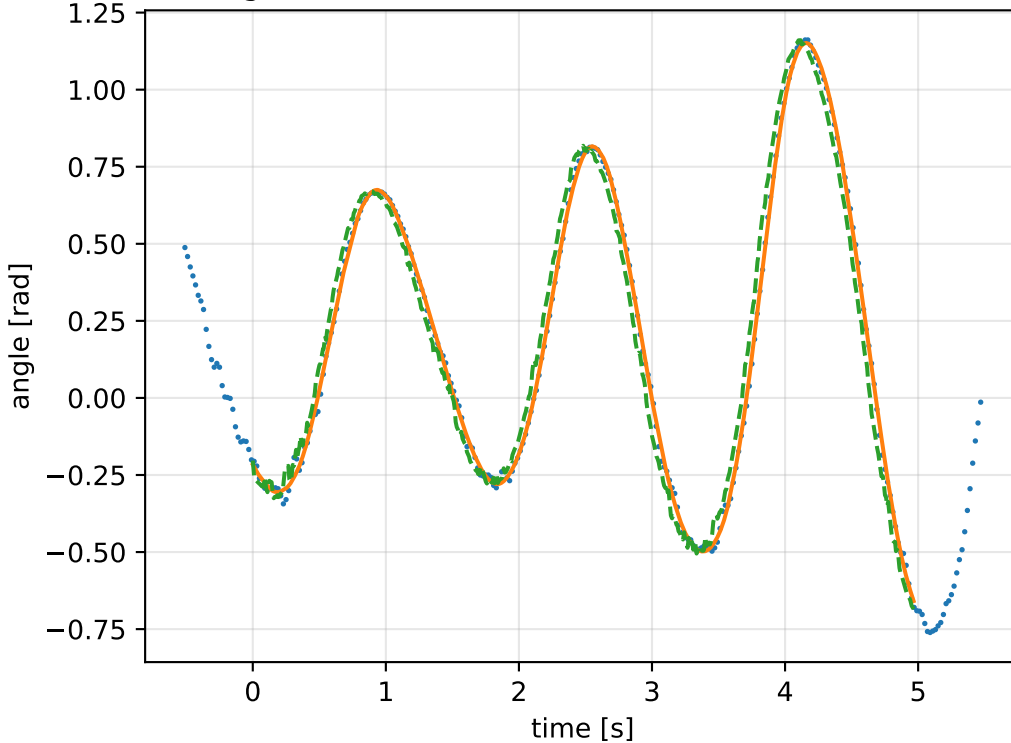
gimbal 1: RMSE=0.0759, max=0.2095 rad



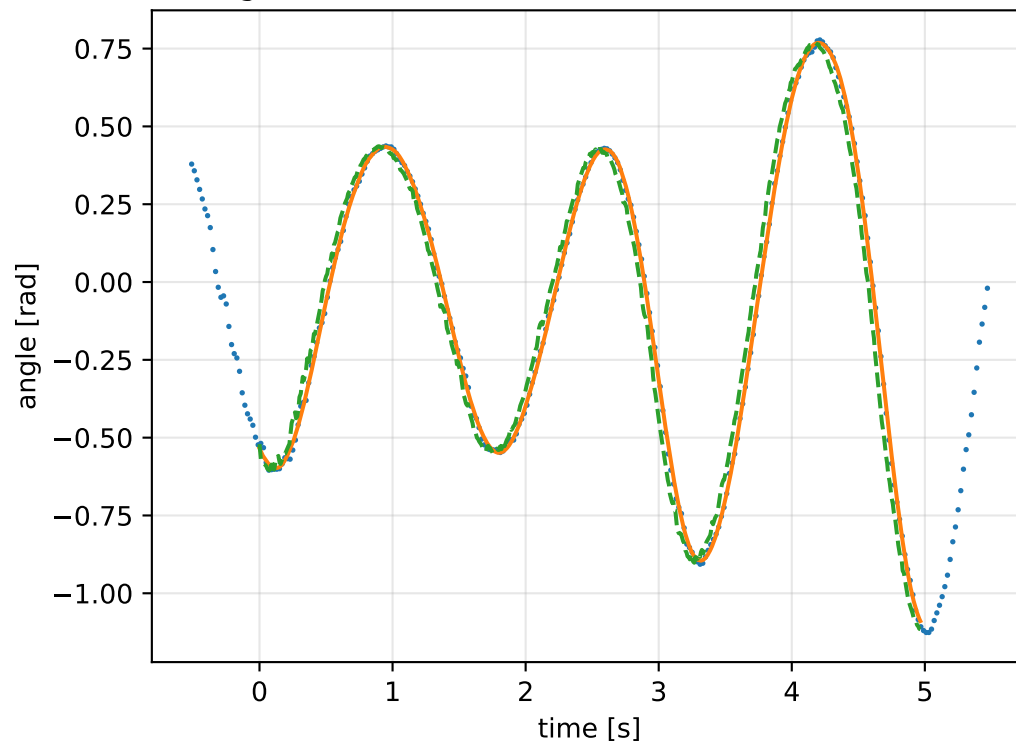
gimbal 2: RMSE=0.08099, max=0.2013 rad



gimbal 3: RMSE=0.07393, max=0.172 rad

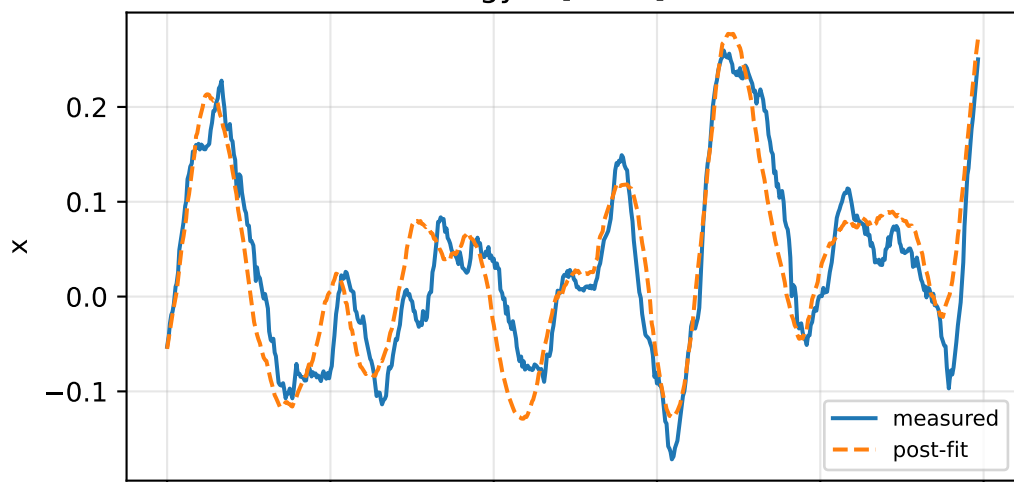


gimbal 4: RMSE=0.07131, max=0.1737 rad

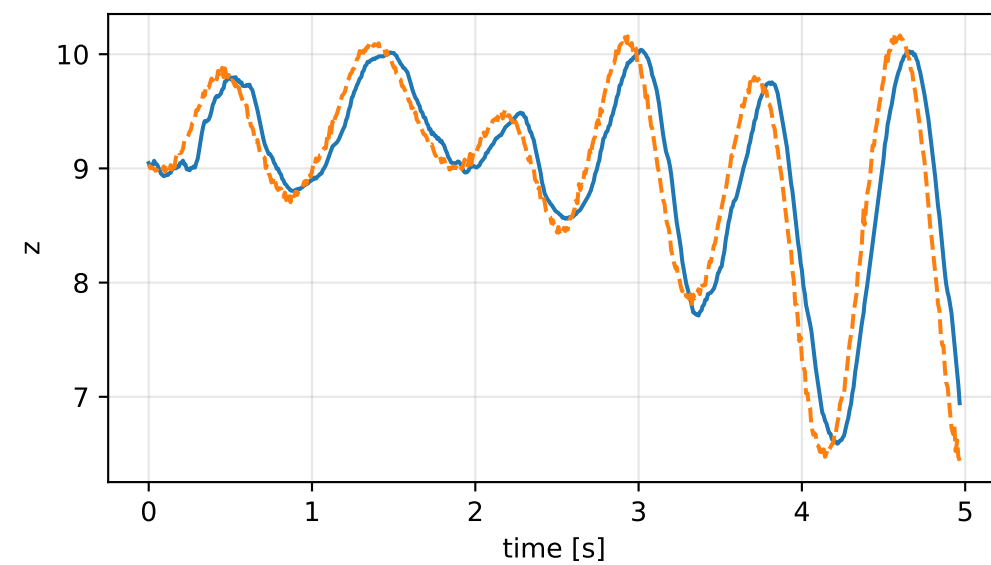
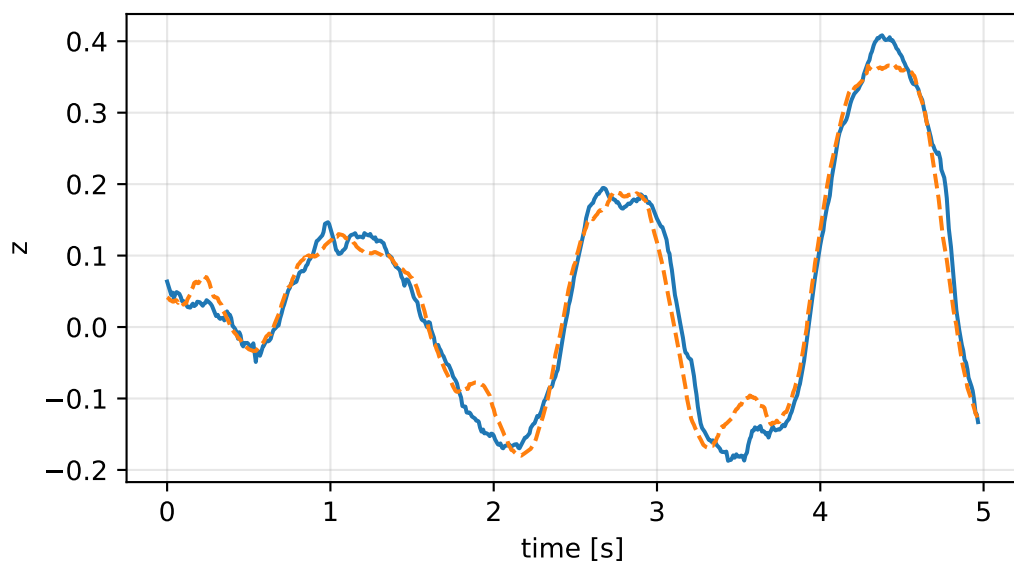
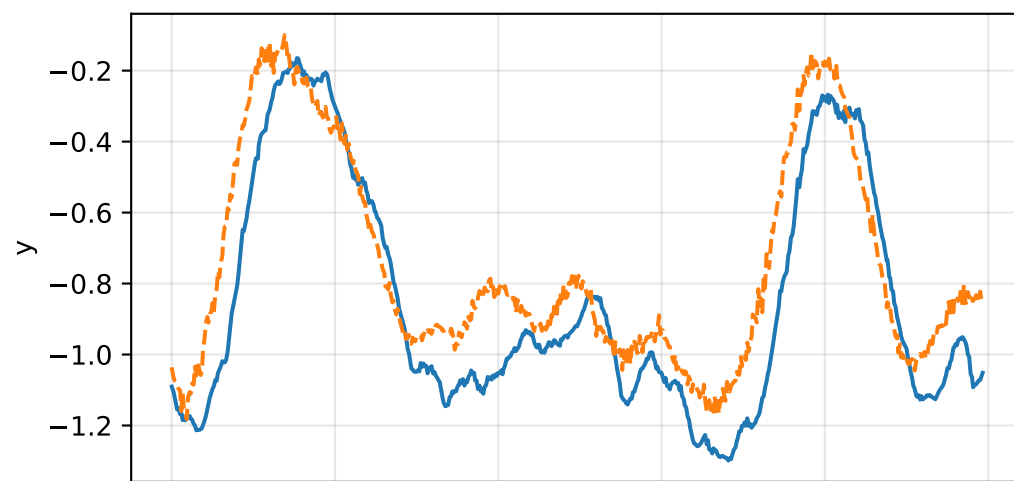
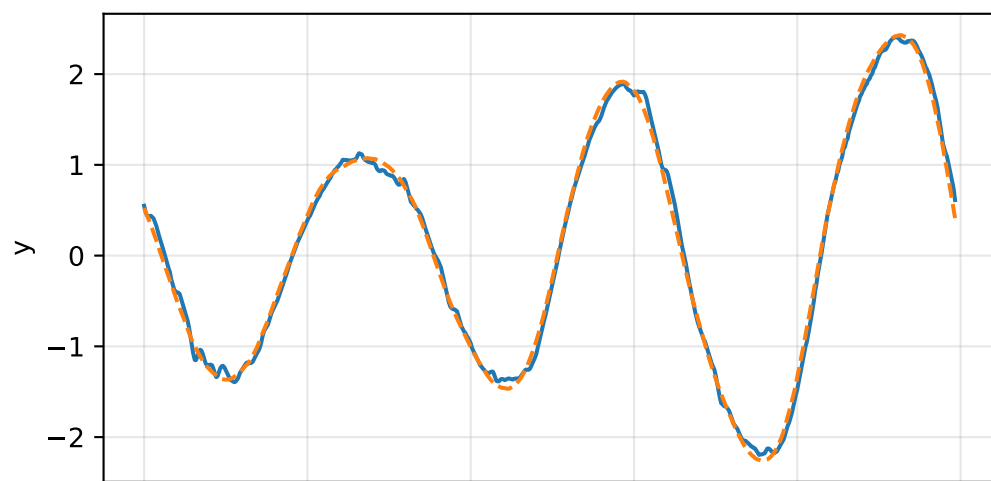
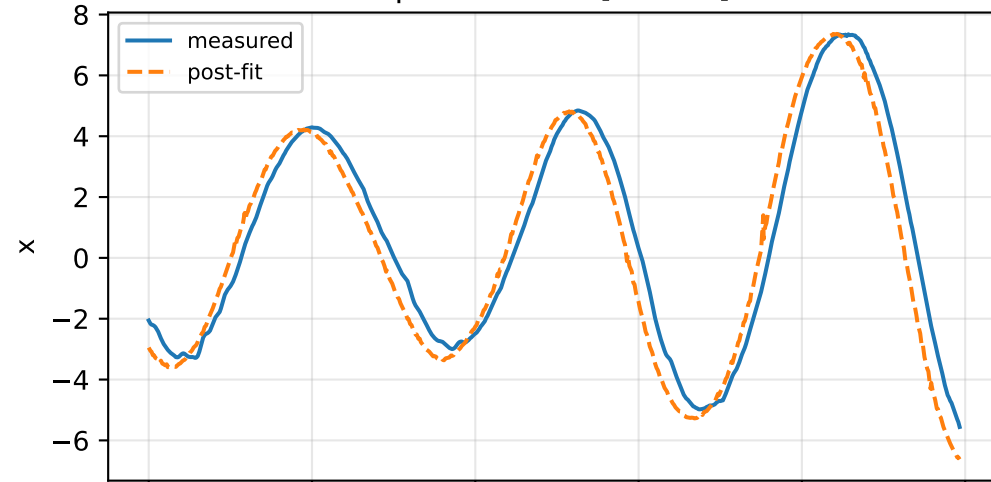


prior_free: IMU comparison (diagnostic only)

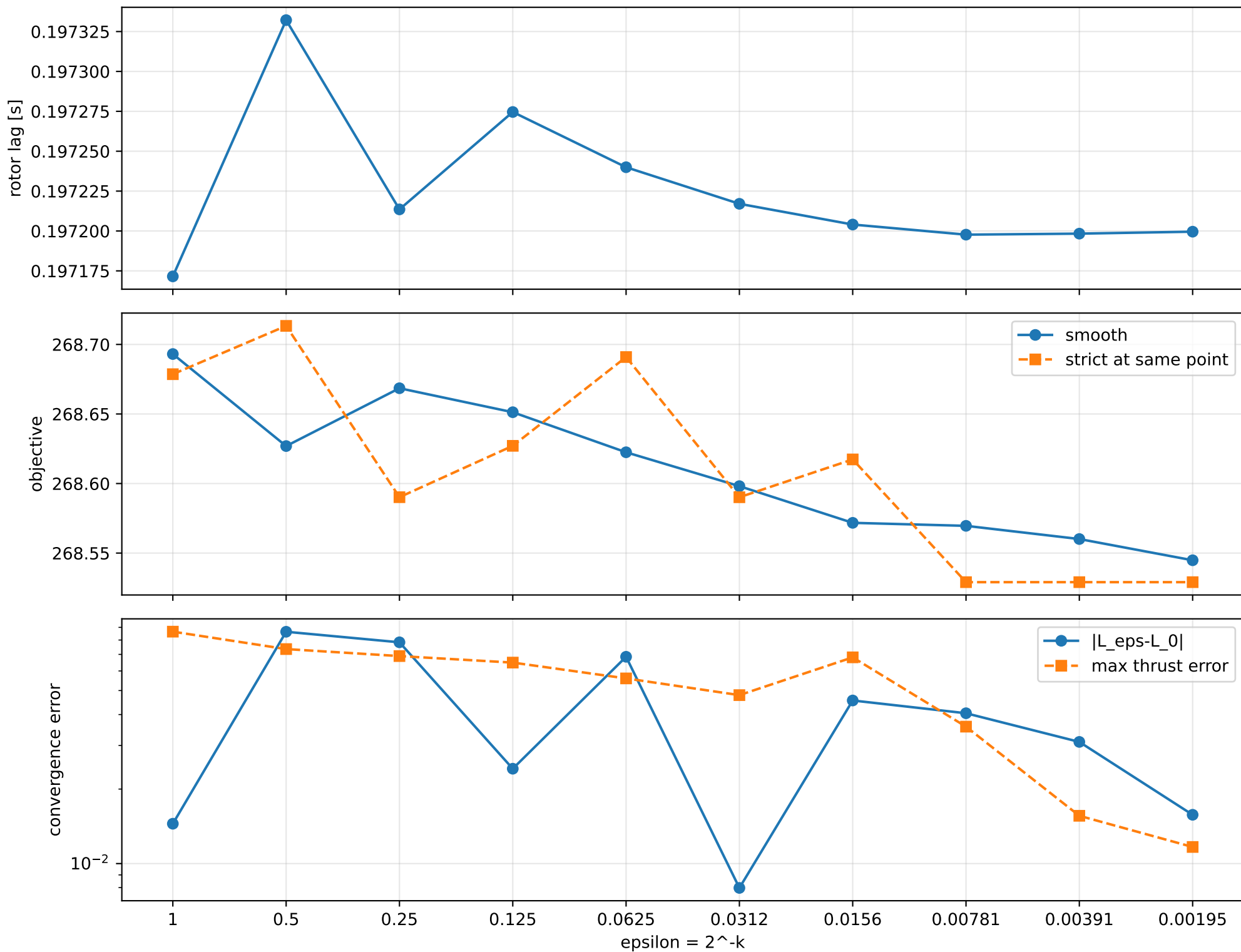
gyro [rad/s]



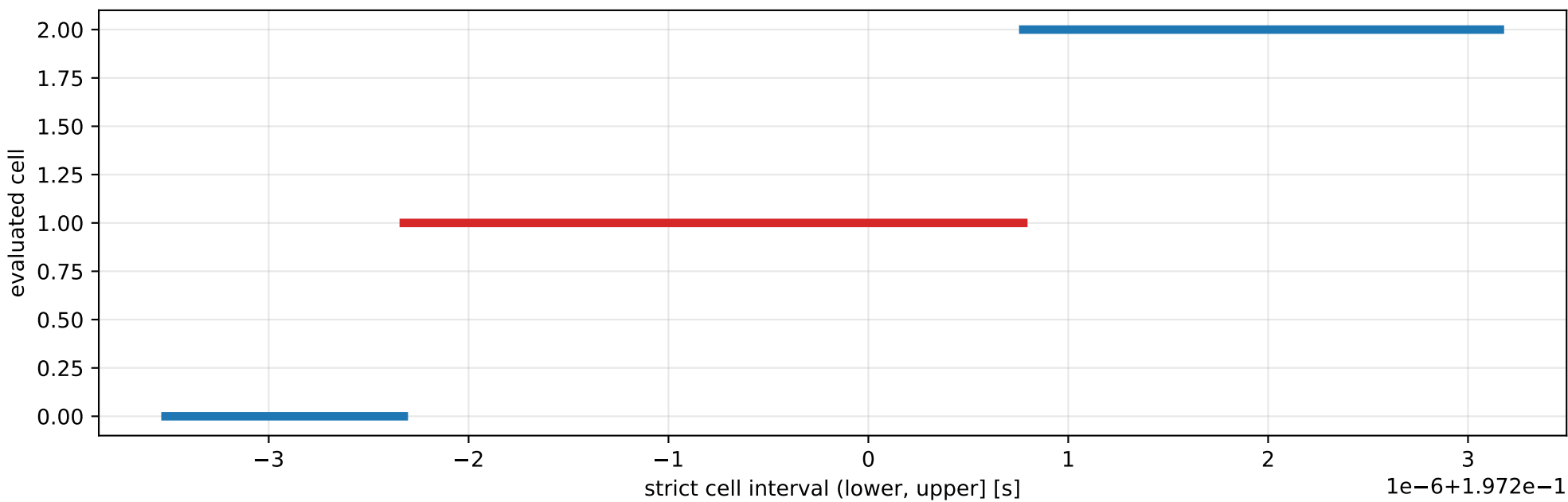
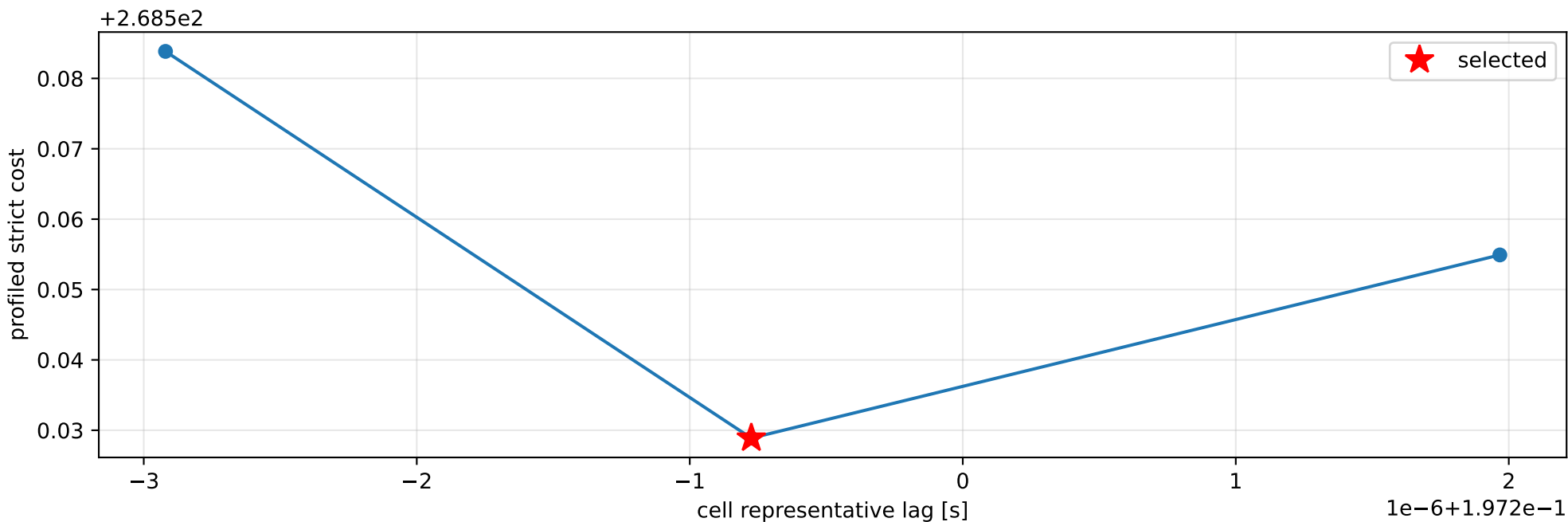
specific force [m/s^2]



prior_free: smooth-to-strict continuation



prior_free: exact strict-ZOH lag cells



selected (0.197197676, 0.197200775] s; final smooth 0.197199564 s; data boundary=False

prior_free: parameter estimate

Case: prior_free

status: completed (all stages completed)

mass [kg]: nominal 2.35159759; estimated 3.47573919

inertia [kg m²]:

```
[[ 3.53067464e-01 -4.27100120e-04 -5.73203733e-02]
 [-4.27100120e-04  2.68867277e-01  3.30980041e-05]
 [-5.73203733e-02  3.30980041e-05  5.96255973e-01]]
```

CoG body [m]: [0.01762808 -0.00411735 -0.04437562]

force effectiveness: [1.16405654 1.49895804 1.30895698 1.08525949]

scale-free J/m [m²]:

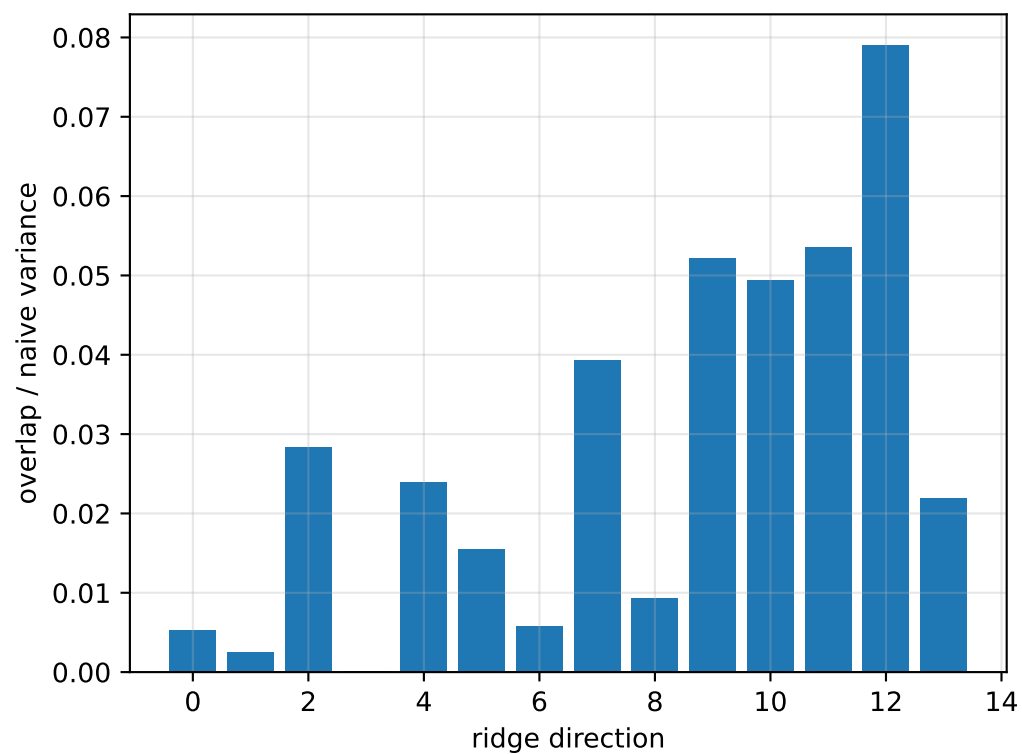
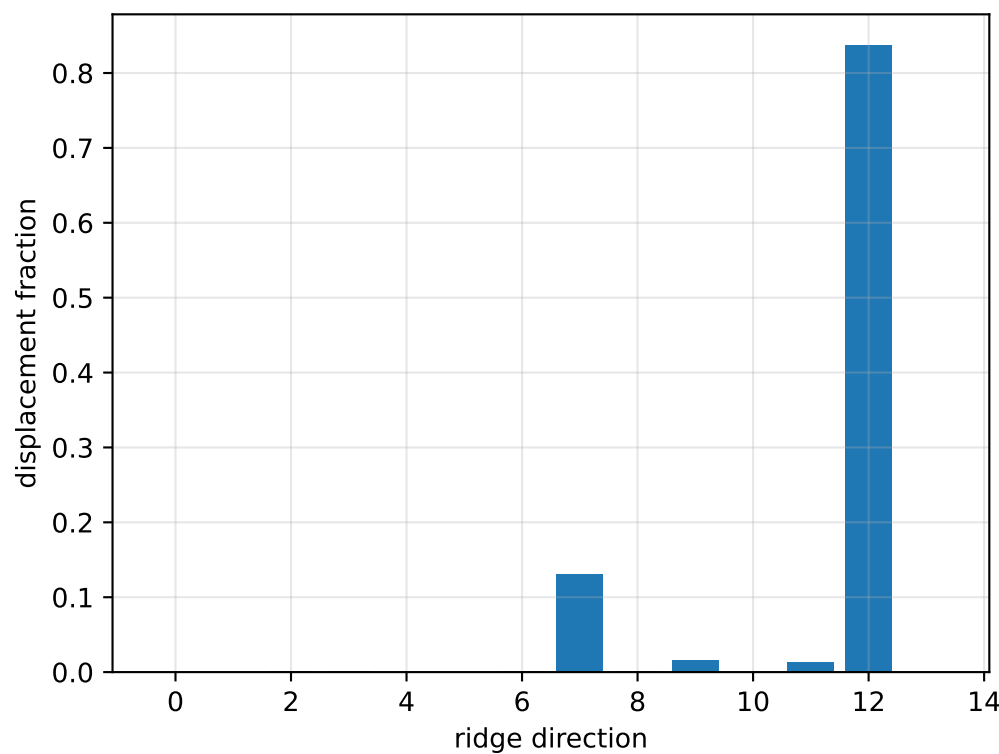
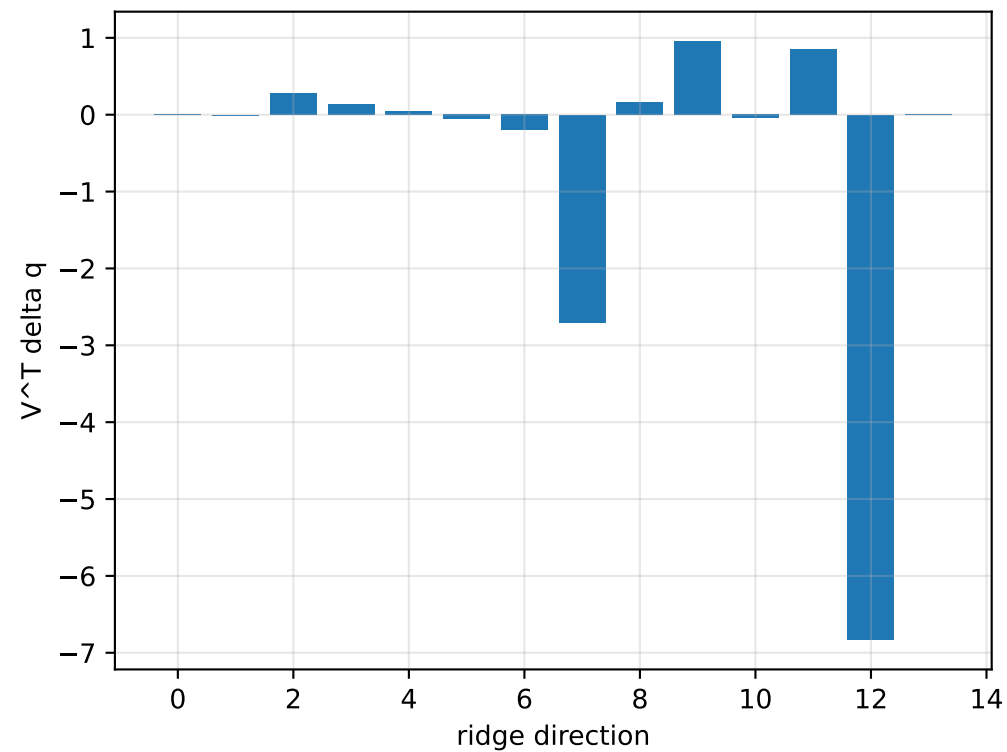
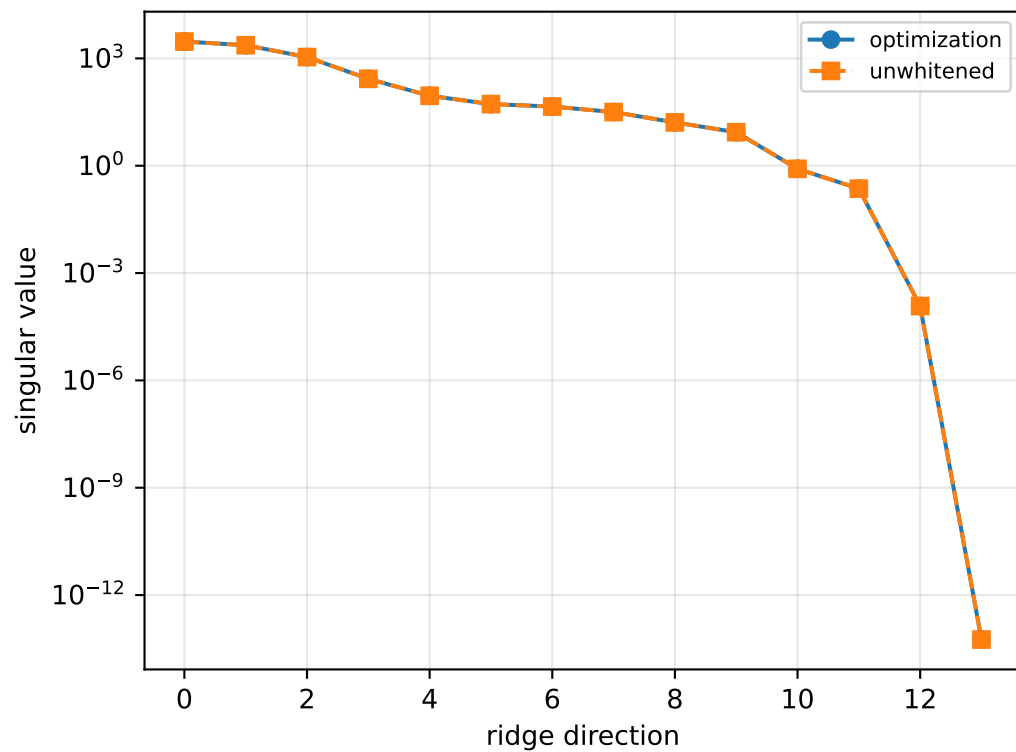
```
[[ 1.01580540e-01 -1.22880371e-04 -1.64915634e-02]
 [-1.22880371e-04  7.73554236e-02  9.52257991e-06]
 [-1.64915634e-02  9.52257991e-06  1.71547962e-01]]
```

scale-free f/m: [0.33490906 0.43126309 0.37659816 0.31223847]

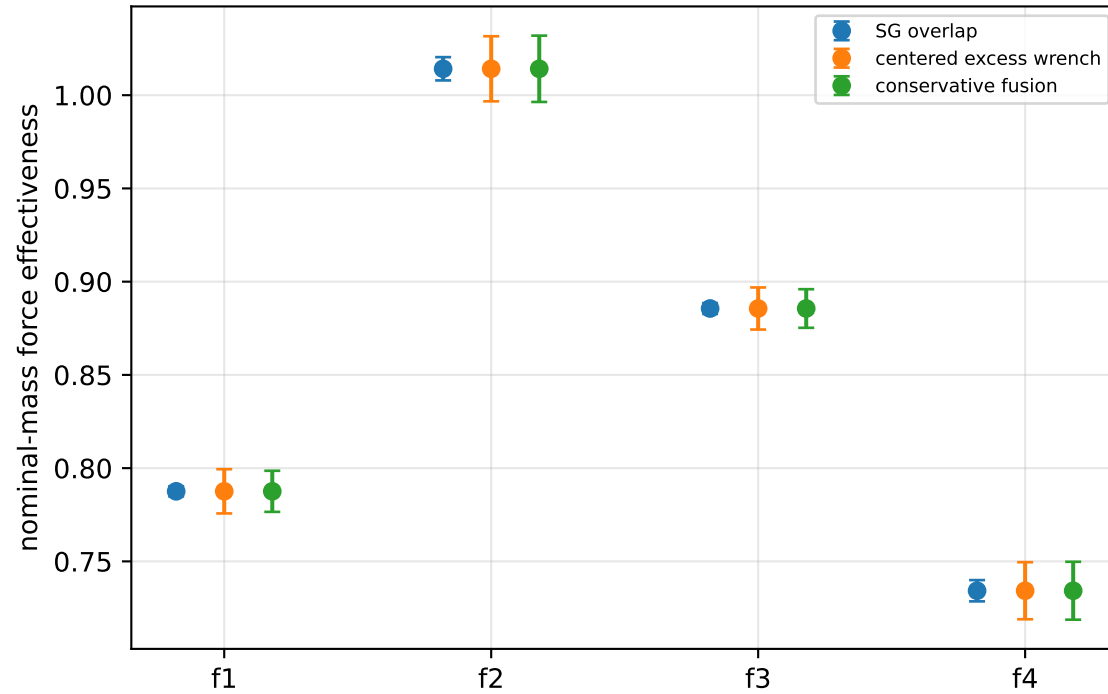
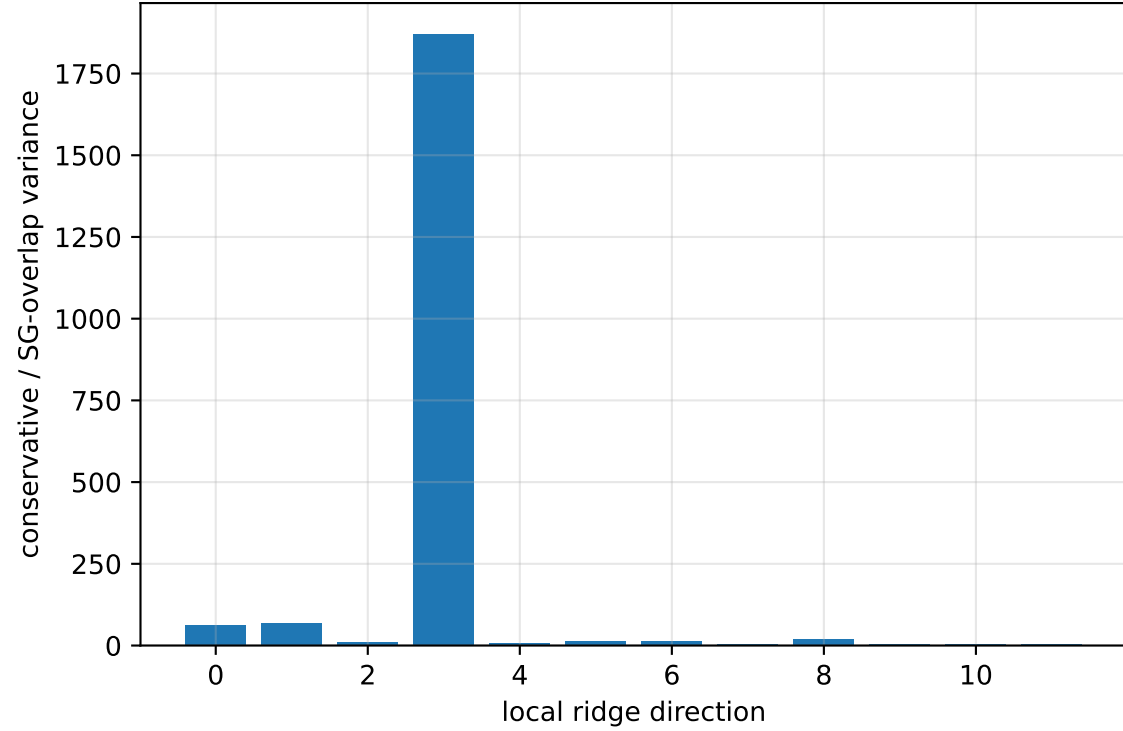
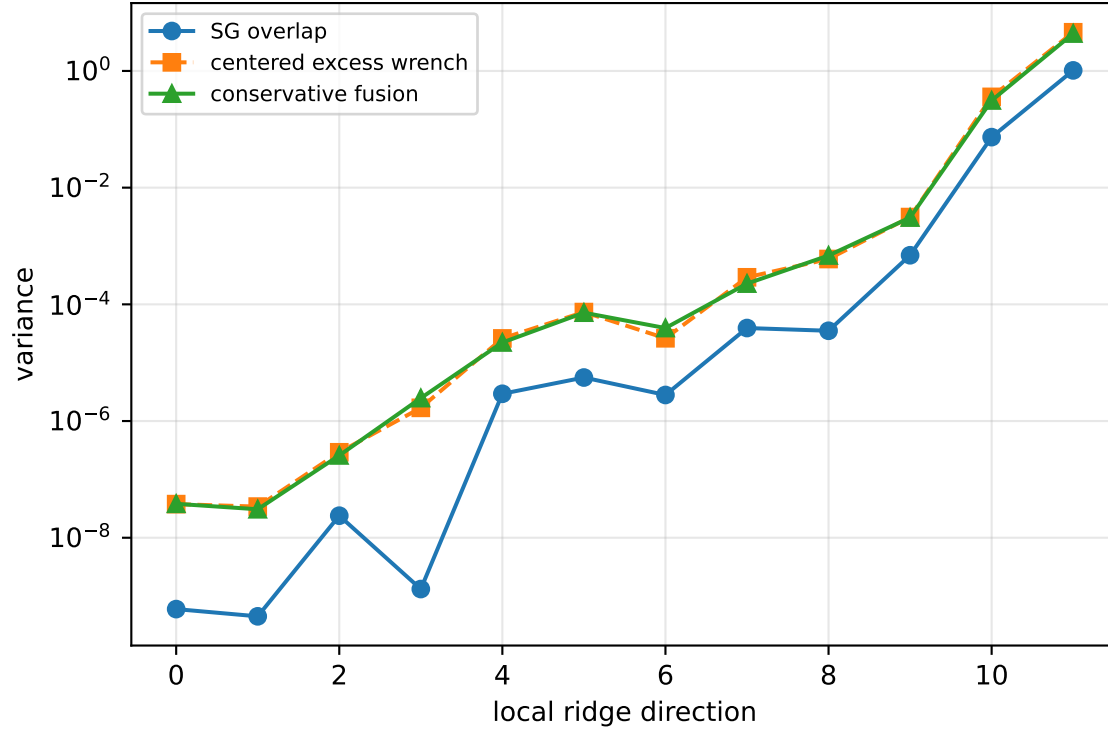
rotor lag cell: (0.197197676, 0.197200775] s

representative: 0.197199225 s

prior_free: ridge spectrum (rank 13, nullity 1, ||Jv||=1.64e-13)



prior_free: Conservative fusion uncertainty



Top three non-gauge ambiguous directions
 $\text{tr}(\mathbf{M}_{\text{res}}) / \text{tr}(\mathbf{M}_{\text{SG}}) = 44.2$
wrench-to-acceleration recovery max error = $3.77\text{e-}15$
exact scale gauge ridge index = 13 (alignment 1)

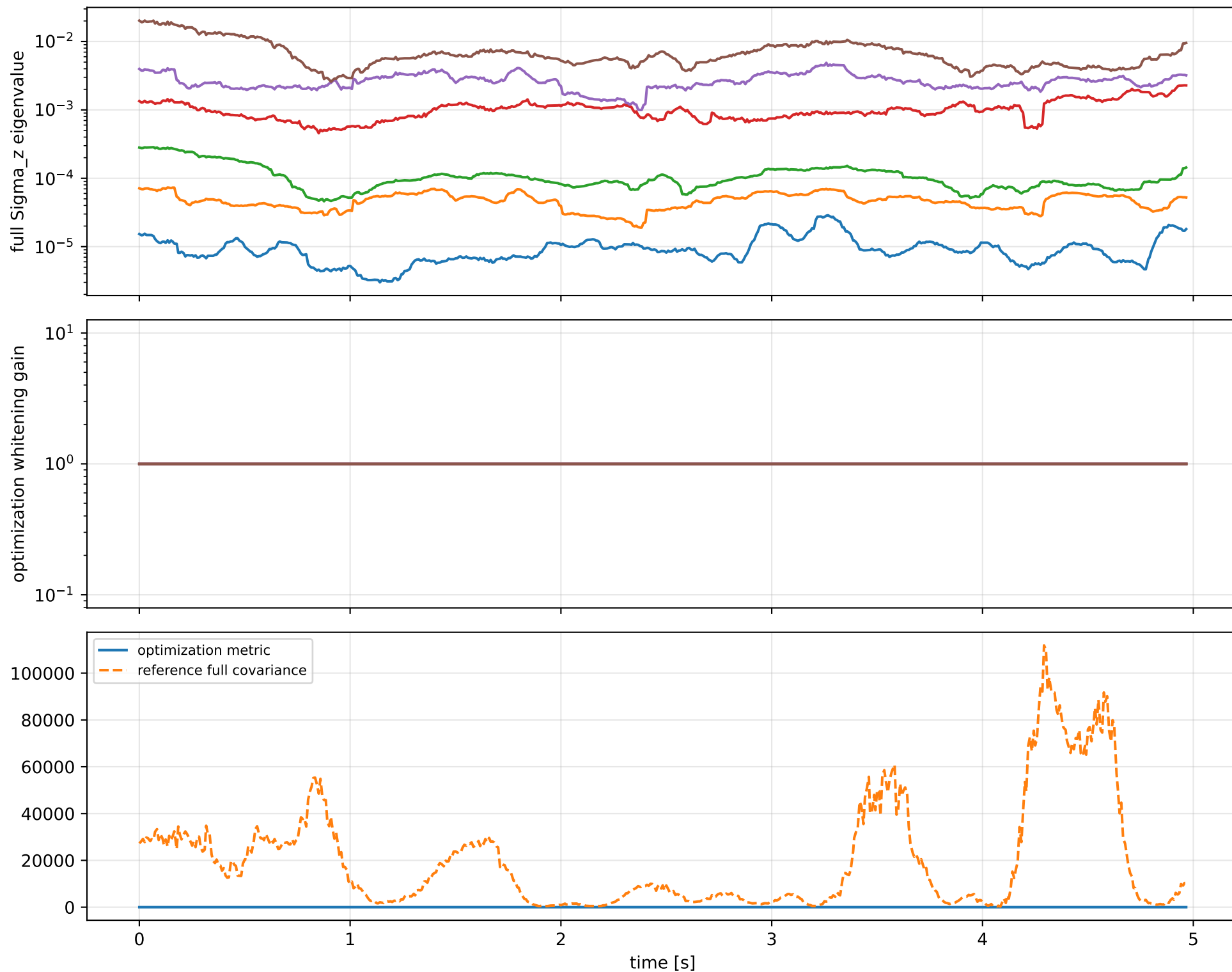
index 11: variance=4.37, ratio=4.275
 $\log_{\text{mass}}=-0.0479$; second_moment_diag_1=-0.415
second_moment_diag_2=-0.0463; second_moment_diag_3=+0.701
second_moment_offdiag_12=+0.0441; second_moment_offdiag_13=-0.567
second_moment_offdiag_23=-0.0268; cog_x=+6.41e-06
cog_y=-0.000197; cog_z=+2.07e-05
log_force_eff_1=-0.0475; log_force_eff_2=-0.0469
log_force_eff_3=-0.049; log_force_eff_4=-0.0484

index 10: variance=0.3116, ratio=4.244
 $\log_{\text{mass}}=+0.00245$; second_moment_diag_1=+0.0214
second_moment_diag_2=+0.0107; second_moment_diag_3=-0.0402
second_moment_offdiag_12=+0.884; second_moment_offdiag_13=+0.0239
second_moment_offdiag_23=-0.463; cog_x=+0.00142
cog_y=-0.00354; cog_z=+0.000339
log_force_eff_1=+0.0105; log_force_eff_2=+0.0204
log_force_eff_3=-0.00517; log_force_eff_4=-0.0202

index 9: variance=0.003069, ratio=4.423
 $\log_{\text{mass}}=-0.148$; second_moment_diag_1=-0.034
second_moment_diag_2=+0.899; second_moment_diag_3=-0.0816
second_moment_offdiag_12=-0.00946; second_moment_offdiag_13=-0.0844
second_moment_offdiag_23=+0.0136; cog_x=+0.0221
cog_y=-0.0136; cog_z=-0.00395
log_force_eff_1=-0.187; log_force_eff_2=-0.0222
log_force_eff_3=-0.094; log_force_eff_4=-0.332

The conservative fusion covariance deliberately retains the existing SG-overlap uncertainty and adds the uncentered empirical residual score second moment without subtracting the SG contribution. It is intended as a conservative fusion distribution, not as a calibrated generative noise covariance.

prior_free: optimization vs reference covariance



prior_free: wrench / closure (force max 1.29e-14, torque max 6.59e-16)

